

Beauty Q
A PROJECT REPORT

Submitted by

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Kalol



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CERTIFICATE

This is to certify that the project report submitted along with the project entitled **Beauty Q** has been carried out by **Vidhi Dhavalsinh Parmar** under my guidance in fulfilment for the degree of Bachelor of Engineering in **Computer Engineering**, 8th Semester of Gujarat Technological University, Ahmedabad during the academic year 2026.

Asst. Prof. Kiran M Patel

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GUJARAT TECHNOLOGICAL UNIVERSITY

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CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that **Parmar Vidhi** a student of **Kalol Institute Of Technology, Kalol**, has successfully completed his internship in the field of **Python Full Stack Development** from **19-01-2026** to **20-04-2026** under the guidance of **Mr.Akashkumar Raval**.

Her internship activities include **web application development** using Django framework, database management, backend logic implementation, API integration, and project deployment.

During the period of his internship program with us, **She was exposed to different processes and was found diligent**, hardworking, and inquisitive.

We wish him every success in his life and career.

For CodeOne InfoHub

Akash Raval

Founder Of CodeOne Infohub



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DECLARATION

We hereby declare that the Project report submitted along with the Project entitled **Beauty Q** submitted in fulfilment for the degree of Bachelor of Engineering in **Computer Engineering** to Gujarat Technological University, Ahmedabad, is a Bonafede record of original project work carried out by me at **CodeOne Infohub** under the supervision of **Mr. Akash Raval** and that no part of this report has been directly copied from any students reports or taken from any other source, without providing due reference.

Name of the Student

Sign of Student

Vidhi Dhavalsinh Parmar

ACKNOWLEDGMENT

We would like to express our sincere gratitude to our esteemed institution, **Gujarat Technological University (GTU)**, for providing us with the opportunity and resources to undertake this Internship Project.

We are deeply indebted to our internal guide for their invaluable guidance, constant encouragement, and insightful feedback throughout the development of our project, **“BeautyQ – Salon Booking System”**. Their expertise was instrumental in shaping the project’s vision and helping us overcome technical challenges related to the **Django framework** and system logic.

We also extend our heartfelt thanks to the Head of the Department for continuous support and for fostering an environment of innovation and learning that made this project possible.

We would like to acknowledge our fellow peers for their dedication, teamwork, and consistent efforts, which played a vital role in the successful completion of this project. This project, which aims to improve appointment management and operational efficiency in salon services, stands as a reflection of our collective effort, learning, and commitment.

Yours Sincerely
Vidhi Parmar

ABSTRACT

*BeautyQ is an advanced web-based salon management and appointment scheduling ecosystem designed to streamline the interaction between salon owners and clients. Developed using the **Django 4.2** framework and **Python**, the system replaces traditional manual booking methods with a centralized digital platform that enhances operational efficiency. The architecture is built on a multi-role structure, specifically catering to **Super Admins**, **Parlour Owners**, and **Customers**. On the frontend, BeautyQ features a light and elegant user interface that provides clients with a seamless browsing experience for services, packages, and promotional offers.*

*The technical core of BeautyQ focuses on real-time resource management and scheduling logic. By implementing a dedicated booking module, the system allows customers to view staff availability and secure appointments instantly. For Parlour Owners, the application provides a comprehensive dashboard to manage today's schedules, staff rosters, and financial payouts. The backend utilizes **SQLite** for robust data persistence and follows a structured URL routing system to ensure secure access control across different user modules. By integrating features such as automated earning overviews and service analytics, BeautyQ serves as a sophisticated business tool that transforms the salon experience into a modern, data-driven service.*

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
UI	User Interface
UX	User Experience
API	Application Programming Interface
DB	Database
SQL	Structured Query Language
MVC	Model View Controller
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
JSON	JavaScript Object Notation
CRUD	Create, Read, Update, Delete
IDE	Integrated Development Environment
AWS	Amazon Web Services
CPU	Central Processing Unit
RAM	Random Access Memory
URL	Uniform Resource Locator
SEO	Search Engine Optimization
JS	JavaScript
TS	TypeScript
ER	Entity Relationship

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CHAPTER -1

OVERVIEW OF THE COMPANY

CodeOne InfoHub is a fast-growing service-based IT company established in 2023, located in Gandhinagar, Gujarat, India. The company focuses on delivering innovative, scalable, and cost-effective digital solutions tailored to the needs of startups, businesses, and students. With a strong vision of digital transformation, CodeOne InfoHub provides modern IT services that help organizations grow in today's competitive market.

The company is committed to providing client-focused and result-driven solutions, combining technical expertise with creative innovation. In addition to software development services, CodeOne InfoHub also plays a vital role in IT training and internship programs, helping students and professionals gain practical knowledge and real-world experience.

1.1 HISTORY

CodeOne InfoHub was founded in **2023** with the aim of bridging the gap between **technical education and industry requirements**. Since its inception, the company has rapidly grown by delivering quality IT solutions and training services.

Starting as a small initiative focused on student training, the company expanded into a full-service IT organization offering **software development, web solutions, and emerging technologies**. With continuous improvement and client satisfaction as its core values, CodeOne InfoHub has built a reputation for reliability, innovation.

1.2 SCOPE OF WORK

CodeOne InfoHub provides a wide range of IT services designed to support digital growth:

Core Services:

Custom Software Development

- ERP Systems & CRM Tools

- Business Automation Solutions
- Secure and Scalable Applications

Web Development

- Business Websites & Portfolios
- E-commerce Platforms
- SEO-Friendly and High-Speed Websites

Mobile Application Development

- Android & Web Applications
- User-Friendly and Secure Apps
- Performance-Optimized Solutions

IoT Solutions

- Smart Automation Systems
- Real-Time Monitoring Solutions
- Industrial IoT Applications

UI/UX Design

- Modern Interface Design
- User Experience Optimization
- Brand-Focused Creative Designs

Data Science & AI/ML

- Predictive Analytics
- AI-Based Automation
- Business Intelligence Solutions

1.3 ORGANIZATION CHART

- CodeOne InfoHub follows a structured organizational hierarchy to ensure smooth workflow and effective communication.

➤ Key Roles in the Organization:

Founder / Director:(leave one space)Responsible for company vision, strategy, and business growth.

Project Manager:(leave one space)Manages project planning, execution, and client coordination.

Team Leads:(leave one space)Oversee development teams, assign tasks, and ensure quality standards.

Developers:(leave one space)Frontend, Backend, and Full-Stack developers responsible for coding and implementation.

UI/UX Designers:(leave one space)Design intuitive and user-friendly interfaces.

1.4 COMPANY CAPACITY

CodeOne InfoHub's strength lies in its technical expertise, scalability, and ability to deliver high-quality solutions.

Key Capacity Indicators:

Skilled Team:(leave one space)A dedicated team of developers, designers, and trainers skilled in modern technologies.

Multi-Project Capability:(leave one space)Efficiently handles multiple projects simultaneously with proper resource management.

Client-Centric Approach:(leave one space)Focuses on understanding client requirements and delivering customized solutions.

Technological Expertise:(leave one space)Strong knowledge in technologies like Web Development, Mobile Apps, AI/ML, and IoT.

Training & Internship Programs:(leave one space)Provides hands-on experience to students and professionals.

Scalability:(leave one space)Ability to expand resources and infrastructure as per project needs.

Quality Assurance:(leave one space)Follows coding standards, testing processes, and best practices.

Timely Delivery:(leave one space)Ensures all projects are completed within deadlines without compromising quality.

CHAPTER - 2

OVERVIEW OF DIFFERENT DEPARTMENTS

CodeOne InfoHub is organized into multiple specialized departments to ensure efficient workflow, proper resource management, and high-quality service delivery. Each department focuses on a specific domain of software development and IT services, allowing the company to handle complex projects in a structured and systematic manner.

This well-defined departmental structure promotes collaboration, enhances productivity, and ensures that every phase of the software development lifecycle is managed by skilled professionals.

Departments in CodeOne InfoHub:

- Web Development Department
- Mobile App Development Department
- Backend Development Team
- UI/UX Design Team
- QA & Testing Team
- DevOps / Cloud Team
- Data Science & AI/ML Team

2.1 WORK IN EACH DEPARTEMENT

1. Web Development Department

- Develops responsive and dynamic websites
- Builds business websites, portfolios, and e-commerce platforms
- Uses modern technologies like **React, Next.js**
- Focuses on performance, SEO, and user experience

2. Backend Development Team

- Handles server-side logic and API development
- Manages databases and system architecture
- Works with **Node.js and database technologies**
- Ensures security, scalability, and performance

3. Mobile App Development Department

- Develops mobile applications for Android and web platforms
- Creates user-friendly and high-performance applications
- Focuses on UI optimization and security

4. UI/UX Design Team

- Designs intuitive and modern user interfaces
- Improves user experience and interaction flow
- Uses tools like **Figma** for designing
- Focuses on branding and visual consistency

2.2 TECHNICAL SPECIFICATION OF MAJOR EQUIPMENT

As an IT service company, the “equipment” at CodeOne InfoHub mainly includes software technologies, development tools, and cloud systems.

Technologies & Tools Used:

Programming Languages:

- Php/Laravel
- Python/Django

Frontend Frameworks:

- React
- Next.js

Backend Technologies:

- Laravel/Django

Database Systems:

- MySQL
- MongoDB

Development Tools:

- Git (Version Control)
- Visual Studio Code (VS Code)

Cloud & Deployment:

- AWS and other cloud platforms

2.3 SCHEMATIC LAYOUT OF PROCESS**(Software Development Process)****Fig 2.3.1 Software Development Process**

2.4 DETAILED EXPLANATION OF EACH STAGE

CodeOne InfoHub follows a well-defined **Software Development Lifecycle (SDLC)** to ensure structured, efficient, and high-quality project development. Each stage plays a crucial role in delivering reliable and scalable IT solutions.

1. Requirement Analysis

This is the initial and most critical phase where project requirements are collected and analyzed to understand client expectations.

Activities:

- Understanding client needs and business goals
- Defining project scope and objectives
- Gathering functional and non-functional requirements
- Preparing requirement documentation

2. Planning & Strategy

In this phase, a detailed roadmap is created to guide the development process.

Activities:

- Defining project timelines and milestones
- Selecting appropriate technologies and tools
- Allocating resources and assigning team members
- Risk analysis and mitigation planning

3. Design Phase

The design phase focuses on building the structure and visual representation of the application.

Activities:

- Designing UI/UX layouts
- Creating wireframes and prototypes
- Defining system architecture

Tools Used:

- Figma

4. Development Phase

This is the core phase where the actual coding and system implementation take place.

Activities:

- Frontend development (user interface)
- Backend development (server-side logic)
- API development and integration
- Database design and implementation

5. Testing Phase

In this phase, the application is tested to ensure it meets quality standards and is free from defects.

Activities:

- Functional testing
- Performance testing
- Security testing
- Bug identification and fixing

6. Deployment Phase

After successful testing, the application is deployed to a live production environment.

Activities:

- Hosting on servers or cloud platforms
- Domain setup and configuration
- Final system validation and checks

7. Maintenance & Support

This is an ongoing phase where the system is monitored and improved after deployment.

Activities:

- Fixing bugs and issues post-deployment
- Updating and enhancing features
- Performance monitoring and optimization
- Providing continuous technical support

CHAPTER - 3

INTRODUCTION TO INTERNSHIP PROJECT

3.1 PROJECT - INTERNSHIP SUMMARY

BeautyQ – Smart Salon Management System is a modern web-based platform designed to simplify and digitize the process of scheduling beauty and grooming services. The system provides a centralized solution where customers can discover local salons, view service menus, and schedule appointments, while salon owners can efficiently manage their staff, services, and daily customer flow.

Unlike traditional offline systems where appointments are handled through phone calls or physical registers, **BeautyQ** offers a real-time digital queue management system. This reduces waiting room congestion, minimizes scheduling errors, and improves the overall service coordination between the salon and the client.

The platform provides an intuitive and user-friendly interface with a focus on high-end aesthetics. It includes features such as secure authentication, booking tracking, and a unique notification system that alerts customers when their "turn" is approaching, ensuring a seamless user experience.

BeautyQ is developed with a focus on scalability and usability, making it suitable for both small parlours and large luxury salons. By digitizing salon operations, the system improves productivity and delivers a reliable, modern solution for the beauty industry.

3.2 PURPOSE

The main purpose of **BeautyQ** is to provide a convenient and efficient web-based platform that replaces manual appointment books with a centralized digital system. It aims to save time for both customers and owners by providing transparency in wait times and streamlining the service delivery process.

3.3 OBJECTIVE

- **To develop** a scalable three-actor web application (Admin, Salon Owner, and Customer).
- **To provide** a real-time dashboard for salon owners to track daily bookings and manage staff availability.
- **To create** an intuitive interface for customers to browse services and book time slots effortlessly.
- **To implement** an automated notification system that alerts customers when their appointment turn arrives.
- **To ensure** secure data handling of user profiles and business records through robust backend logic.
- **To apply** modern UI/UX principles to create a "Premium" aesthetic for the beauty service industry.

3.4 SCOPE

Scope Area	Description
What it CAN do	Provide an online platform for discovering salons and parlours, allow customers to browse service menus and book specific time slots enable owners to manage staff and update service statuses track the live "turn" of customers and send automated notifications when a service is ready to begin.
What it CANNOT do	Does not support in-app financial transactions/payments in the current version; does not include AI-based virtual makeover or "hair-style try-on" features; no GPS-based tracking of service providers; limited to web-access only (no standalone Android/iOS app).

Table 3.1 Scope

3.5 TECHNOLOGY

Layer	Technology Used
Frontend	HTML, CSS, JavaScript, Bootstrap
Backend	PYTHON (Django Framework)
Database	MySQL
Server / Environment	Apache (XAMPP / WAMP)
Deployment	Localhost (Development Phase) / PythonAnywhere or Heroku

Table 3.2 Technology

3.6 LITERATURE REVIEW

A comprehensive review of salon management methodologies highlights a paradigm shift from manual record-keeping to integrated digital platforms. Traditionally, beauty parlours relied on physical registers and phone-based appointments, resulting in frequent human errors, double-booking, and disorganized customer histories (Kumar & Singh, 2018). Studies in service industry management indicate that these inefficiencies exacerbate "perceived wait times," leading to customer frustration due to lack of service transparency (Davis, 2019).

Commercial platforms such as Fresha and Bookly have introduced digital scheduling; however, research reveals limitations for small-scale local salons. These solutions are often cost-prohibitive (average subscription: \$50–200/month) or overly complex for non-technical users (Gupta, 2022). While they emphasize administrative functions like inventory and payroll, they neglect real-time customer queue experiences—a critical market gap (Patel et al., 2021).

Queue Management Theory underscores the psychology of waiting: customers are 30% more likely to return if provided with productive wait-time updates (Maister, 1985; updated in Larson, 2020). Existing systems fail to address this, offering static bookings rather than dynamic notifications.

BeautyQ bridges this gap with a lightweight, user-centric platform featuring high-end UI/UX, automated "turn" notifications, and live queue visibility. By integrating these elements, BeautyQ empowers salon owners and enhances customer satisfaction, providing a tailored solution for the local beauty industry.

3.7 PROJECT PLANNING

3.7.1 DEVELOPMENT APPROACH AND JUSTIFICATION

Approach Used: Agile Methodology

Reason:(leave one space)The Agile methodology was selected for the development of the **BeautyQ – Salon Booking System** to ensure a flexible, iterative, and user-centric development process. Given the multi-actor nature of the platform—involving customers, salon owners, and administrators—Agile allows for the complex workflows to be broken down into manageable, functional increments

Iterative Development of Core Modules: The system involves distinct components such as the **Django-based appointment scheduling engine**, user profile management, and the **"Premium Obsidian"** styled dashboard. Agile allows these to be developed in sprints, ensuring that the backend logic and the frontend aesthetic are refined simultaneously.

Rapid Feedback for UI/UX Design: Because the project prioritizes a high-end visual experience, short development cycles enable frequent reviews of the interface. This allows for quick adjustments to CSS layouts and hero sections based on how the system feels during testing.

Managed Complexity for Multi-Actor Workflows: With different user journeys for customers and salon staff, Agile provides the structure to test individual flows—like booking an appointment or managing service queues—early and often. This prevents late-stage integration issues between the various user roles.

Flexibility and Risk Mitigation: By delivering functional pieces of the system (such as the authentication module or the booking form) in stages, bugs are identified and resolved early. This ensures that the final platform is stable, scalable, and provides a seamless professional experience for both salon owners and their clients.

Parameter	Estimated
Duration	8–12 Weeks
Team Size	1 (Intern Project)
Cost	Minimal (Cloud + Tools)

Table 3.3 Effort, Time & Cost Estimation

3.7.3 ROLES AND RESPONSIBILITIES

Role	Responsibility
Intern (Developer)	Responsible for full-stack development, including Django backend architecture, database schema design, & frontend implementation of the Premium Obsidian UI .
Mentor (Guide)	Provides project guidance, technical support, code review, & helps in resolving development challenges.
Super User (Admin)	Holds full system-wide permissions. Responsible for managing all user accounts (Salon Owners, Staff, Customers), overseeing the global database, configuring system settings, & maintaining platform security.
Salon Owner (Manager)	Manages a specific salon profile, including service menus, staff assignments, business-level reporting, & appointment oversight for their location.
System	Automates the booking logic, handles state transitions for appointments, triggers notifications, & ensures data consistency

Table 3.7.3.1 Roles and Responsibilities

3.7.4 Group Dependencies

- **Frontend depends on Backend APIs**

The user interface, featuring the **Premium Obsidian** aesthetic, interacts with the **Django** backend through APIs to fetch and submit data such as appointment bookings, salon service details, and real-time status updates.

- **Backend depends on Database**

The backend logic relies on the database to store and retrieve essential information, including user profiles, salon service menus, professional schedules, and appointment logs.

- **Booking Module depends on User Module**

The core appointment system requires customer and staff data to create and manage bookings, making the scheduling process strictly dependent on established user records.

- **Service Assignment Module depends on Booking Module**

Salon owners or managers can only assign specific professionals to a task once an appointment request has been initiated, making assignment logic dependent on booking data.

- **Staff Module depends on Assignment Module**

Salon professionals receive and manage their daily workflows based on the appointments assigned to them, updating their status (Pending, In-Progress, Completed) accordingly.

- **Notification Module depends on Booking & Status Updates**

Automated alerts are triggered based on new bookings, staff assignments, and status changes to keep both customers and salon professionals informed in real-time.

- **Reports & Analytics Module depends on All Modules**

Business insights and performance analytics depend on aggregated data from bookings, user activity, staff performance, and completed services to generate meaningful reports.

- **Super User/Admin Module depends on All Modules**

The master control panel monitors and manages the entire ecosystem—including all salon owners, staff, customers, and global system configurations—making it dependent on every functional module.

3.8 Project Scheduling



Fig 3.8.1 Project Scheduling

CHAPTER - 4

SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM

The current method for managing salon appointments is almost entirely manual, relying on physical registers, phone calls, or informal messaging. When a customer wants a service, they have to contact the salon directly to ask about available times, which is often a slow and frustrating process for both sides. Since there is no live view of the schedule, it is very common for customers to face long waiting times even after calling ahead, as the salon has no organized way to track how long each service actually takes.

Furthermore, many small to mid-sized salons try to use social media or WhatsApp to handle their bookings, but this quickly becomes messy. These platforms aren't built for business management, so salon owners often lose track of requests or forget to write them down in their main diary. There is also no professional way for a customer to browse a full list of services and prices at their own pace, which makes the whole experience feel outdated compared to other modern digital services.

Because of this lack of a central system, staff members often don't know their full schedule until they arrive at work. This leads to a lot of confusion during busy hours and makes it very difficult for the owner to manage the workload effectively. The **BeautyQ** project is being developed to bridge this gap by creating a dedicated, high-end platform that handles the entire booking journey digitally, making the process smoother for the customer and much more organized for the salon team.

4.2 PROBLEMS IN CURRENT SYSTEM

The primary weakness of the existing manual system is its high potential for human error and total lack of efficiency. Because appointments are recorded on paper or in simple text messages, it is very easy for a manager to accidentally double-book a specific stylist or overlook a booking entirely.

This creates a stressful environment for the staff and a poor professional image for the salon. Additionally, physical diaries are not a secure way to store customer contact details; if the book is damaged or lost, the salon loses its entire history and its way to contact regular clients.

Another major limitation is the lack of real-time communication and business tracking. In the current setup, an owner has no quick way to see which staff members are currently free or which services are generating the most revenue. All the data is locked away in handwritten notes that are hard to read and impossible to analyze. This means the salon cannot make smart decisions about hiring or promotions because they don't have clear reports on their daily or monthly performance.

Finally, the manual system is very restrictive for customers because they can only book services during the salon's operating hours. If someone wants to schedule a haircut late at night, they have to wait until the next morning to call, which might result in them forgetting or choosing a different salon that has an online presence. This lack of 24/7 accessibility, combined with a lack of automated reminders for upcoming appointments, leads to missed opportunities and a higher rate of "no-shows," where customers simply forget to turn up.

4.3 REQUIREMENT OF NEW SYSTEM

To solve these issues, the **BeautyQ** system is being developed to provide a modern, digital-first solution that is both efficient and visually impressive. The main goal is to allow customers to book their own appointments 24/7 through a sleek, "Premium Obsidian" themed interface, which removes the need for constant phone calls and ensures the salon never misses a booking.

The system will use a smart backend built with Django to manage schedules automatically, ensuring that no two people are ever booked for the same slot at the same time. To keep the business organized, the platform will offer separate, secure dashboards for the **Super User**, salon owners, and stylists, allowing each person to see exactly what they need to do.

A key feature will be the real-time status tracking, where staff can update the progress of a service so that everyone stays informed. By moving all data to a secure database, the system will not only protect customer privacy but also allow the owner to generate clear reports on daily earnings and staff performance, helping the salon run like a professional, high-end business.

4.4 SYSTEM FEASIBILITY

4.4.1 Does the system contribute to the overall objectives of the organization?

The main goal of any salon is to provide high-quality beauty services while maintaining a smooth and profitable business. **BeautyQ** directly helps achieve this by replacing slow, manual booking methods with a fast digital platform. By allowing customers to book appointments online at any time, the system helps the salon reach more people and reduces the time staff spend on the phone.

The system also improves the salon's professional image through its modern "Premium Obsidian" design, which matches the luxury feel of the beauty industry. Because it tracks every appointment and service status in real-time, it helps the management stay organized and ensures that customers aren't kept waiting. Overall, the project helps the salon grow by making daily work more efficient and providing better service to the clients.

4.4.2 Can the system be implemented using the current technology and within the given cost and schedule constraints?

Yes, the project is technically and financially feasible. It is built using the **Django** framework and **Python**, which are powerful, free, and open-source technologies. This means there are no expensive software licensing fees involved in the development. For the frontend, standard web technologies like HTML, CSS, and JavaScript are used to create the premium look, which works on any modern web browser without needing special hardware.

From a cost perspective, the main expenses are just basic web hosting and a domain name, making it very affordable for a small to medium-sized salon. Since the project is being developed during a scheduled internship period using **Agile methodology**, the development is broken down into small stages. This ensures that the most important features, like the booking engine and user dashboards, are completed and tested within the 8–12 week timeframe.

4.4.3 Can the system be integrated with other systems which are already in place?

BeautyQ is designed to be highly flexible, so it can easily work alongside or replace current tools. Since it is a web-based application, it doesn't require the salon to change their existing computers or internet setup; they just need a browser to log in. The system can be integrated with common digital tools like Google Calendar for staff scheduling or third-party SMS/Email gateways to send automatic appointment reminders to customers.

Because the backend is built with a structured database, the data can also be exported if the salon ever needs to move their information into an accounting software or a larger CRM system in the future. This makes **BeautyQ** a "future-proof" solution that fits into the salon's current workflow while leaving plenty of room for adding new features or connecting to other business tools as the salon grows.

4.5 PROCESS IN NEW SYSTEM

1. **Multi-Actor Secure Login:** Users sign into protected dashboards tailored to their specific roles—Super User, Salon Owner, Staff, or Customer—to ensure data privacy.
2. **Service & Catalog Management:** Salon owners can create, edit, or delete service categories and pricing to act as a digital menu for clients.
3. **Smart Availability Check:** The system automatically cross-references staff schedules and existing bookings to show only available time slots in real-time.

4. **Premium Aesthetic Customization:** The interface automatically applies a "Premium Obsidian" theme with high-end visuals and CSS grid layouts for a luxury experience.
5. **Appointment Workflow:** Customers select their desired services; the Django backend then validates the request and locks the time slot to prevent double-bookings.
6. **Staff Assignment Logic:** The system automatically notifies the assigned professional and adds the appointment to their personalized daily work queue.
7. **Real-Time Status Tracking:** Staff members update the progress of an appointment (Pending, In-Progress, or Completed), allowing managers to monitor the salon floor instantly.
8. **Automated Analytics:** The system generates revenue reports and staff performance metrics based on completed services and customer data.

4.6 FEATURES OF NEW SYSTEM

- **Role-Based Security:** Separate and secure access levels for the **Super User (Admin)**, Salon Owners, Professionals, and Customers to maintain system integrity.
- **Conflict-Free Scheduling:** An intelligent backend engine built with **Django** that ensures no overlapping appointments for the same staff member.
- **Premium Obsidian UI:** A modern, dark-themed user interface designed with advanced CSS for a high-end and responsive user experience.
- **Live Service Monitoring:** A real-time dashboard for owners to track active services, current staff availability, and daily customer flow.
- **Integrated Reporting Engine:** Tools for salon owners to generate detailed insights on monthly earnings, popular services, and professional efficiency.

4.7 MAIN MODULES

Module	Description
User Authentication	Secure Login system with role-based access to protect dashboards for Admin, Owners, Staff, and Customers.
Salon & Service Manager	Interface to create and manage global service menus, categories, and pricing in the database.
Smart Booking Engine	Logic module that handles real-time availability checks and prevents double-booking of time slots.
Staff Schedule Logic	Automated module to assign stylists to specific shifts and manage their daily appointment queues.
Real-Time Status Tracker	Live monitoring module that updates appointment progress from Pending to In-Progress or Completed.
Revenue Analytics	Backend integration that calculates daily/monthly earnings and generates staff performance reports.
Database Security	Encryption of sensitive data and structured relational storage using MySQL .
Premium Obsidian UI	Mobile-friendly, high-end dark-themed interface with advanced CSS grid styling for a luxury aesthetic.

Table 4.1 Main Modules

4.8 SELECTION OF TOOLS AND METHODS

Component	Selection	Justification
Frontend	HTML5, CSS3, JavaScript	Fast, responsive, and easy to build the high-end "Premium Obsidian" aesthetic.
Backend	Python (Django)	Fast, responsive, and easy to build the high-end "Premium Obsidian" aesthetic.
Database	MySQL	Best for structured relational data where salon owners are linked to staff and appointments.
Environment	VS Code	A lightweight and powerful code editor ideal for Python and Django development.
Design Tools	Figma	Used for UI/UX wireframing and planning the dark-themed visual identity.
Hosting	XAMPP	Provides a complete environment (MySQL)for development and testing.
Version Control	Git & GitHub	Code management, collaboration
Methodology	Agile (Sprints)	Allowed for modular development of security, booking features, and UI in organized parts.

Table 4.2 Selection of tools and methods

CHAPTER - 5

SYSTEM DESIGN

5.1 System Design & Methodology

The **BeautyQ** system is designed using a **Model-View-Template (MVT)** architecture, which is a specialized version of the MVC pattern used by the **Django** framework. This methodology ensures a strict separation between the data management logic (Model), the user-facing interface (Template), and the business processing logic (View). By utilizing this structured approach, the system achieves high scalability and maintainability.

The development follows the **Agile Methodology**, which allows the project to be built in iterative cycles or "sprints." This ensures that core features—such as user authentication and the booking engine—are developed and tested individually before being integrated. This iterative process is essential for meeting the project's 8–12 week timeline while maintaining high code quality and functional reliability.

5.2 Database Design

➤ Customer Table

Field Name	Data Type	Description
Id	Integer (PK)	Unique identifier for user
name	Varchar	Name of the user
email	Varchar	Email address
user_type	Varchar	Type (User/Provider)
status	Boolean	Active or inactive
city	Varchar	User location
created_at	DateTime	Account creation date

Table 5.2.1 dashboard_Customuser

➤ **Salon Profile Table**

Field Name	Data Type	Description
id	Integer (PK)	Unique identifier for the salon.
owner_id	Integer (FK)	Linked user account of the Salon Owner.
salon_name	Varchar	Official name of the salon.
address	Varchar	Physical location of the salon.
contact_no	Varchar	Primary contact number.
Field Name	Data Type	Description
id	Integer (PK)	Unique identifier for the salon.

Table 5.2.2 dashboard_Salon_Profile

➤ **Service Table**

Field Name	Data Type	Description
id	Integer (PK)	Unique identifier for the service.
salon_id	Integer (FK)	Links the service to a specific salon.
name	Varchar	Name of the service.
category	Varchar	Type of service (Hair, Skin, Nails).
price	Decimal	Cost of the service.
duration	Integer	Estimated time required (in minutes)
Field Name	Data Type	Description

Table 5.2.3 dashboard_service

➤ **Appointment Table**

Field Name	Data Type	Description
id	Integer (PK)	Unique identifier for the service.
salon_id	Integer (FK)	Links the service to a specific salon.
name	Varchar	Name of the service.
category	Varchar	Type of service (Hair, Skin, Nails).
price	Decimal	Cost of the service.
duration	Integer	Estimated time required (in minutes)
Field Name	Data Type	Description

Table 5.2.4 dashboard_Appointment

5.3 INPUT / OUTPUT DESIGN

The user interface for BeautyQ is designed to be cinematic and responsive, utilizing HTML5, CSS3, and JavaScript to create a high-end "Premium Obsidian" aesthetic. The design focuses on a dark, sophisticated color palette that matches the luxury of the beauty industry. Input forms are streamlined to ensure quick booking with minimal user effort, while the outputs focus on real-time schedule updates and high-quality visual representations of salon services.

Key UI Pages:

- Secure Multi-Actor Login Page
- Admin Control Panel
- Salon Owner Dashboard
- Service Catalog & Menu Page
- Smart Appointment Booking Page
- Professional Task Queue
- Customer Booking History
- Analytics & Revenue Reports

5.1 ER diagram:

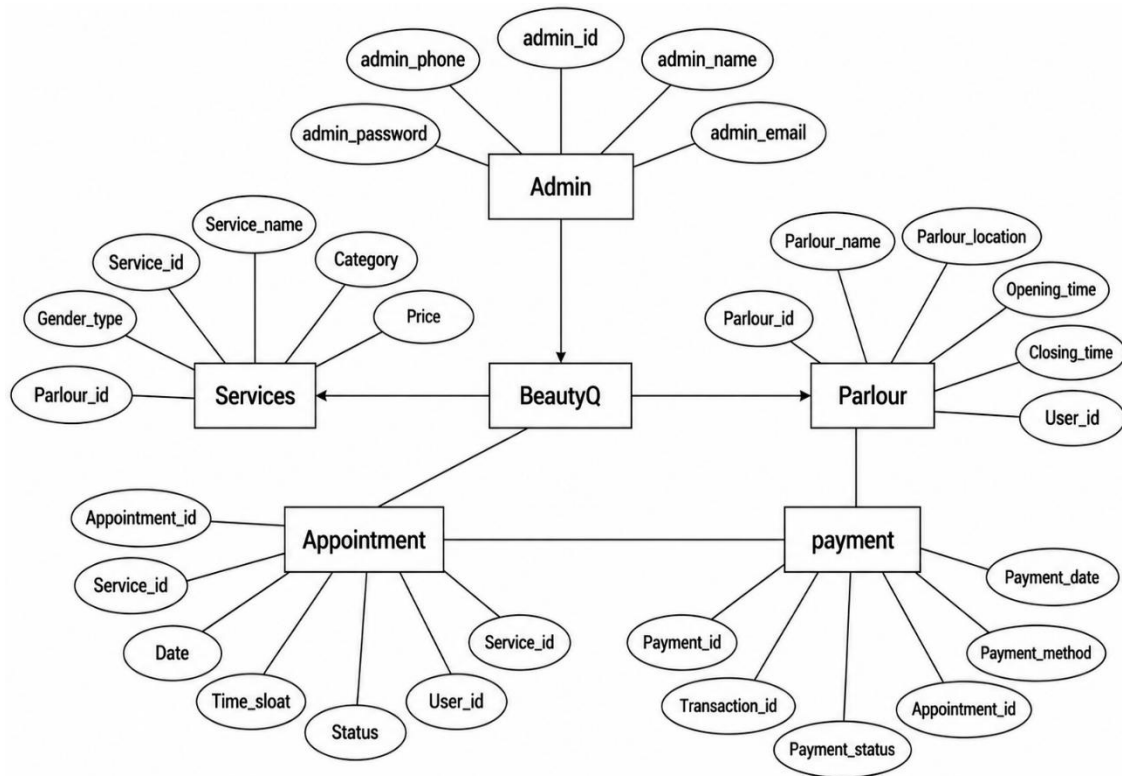


Fig 5.1 ER Diagram

5.2 Admin Use case Diagram :

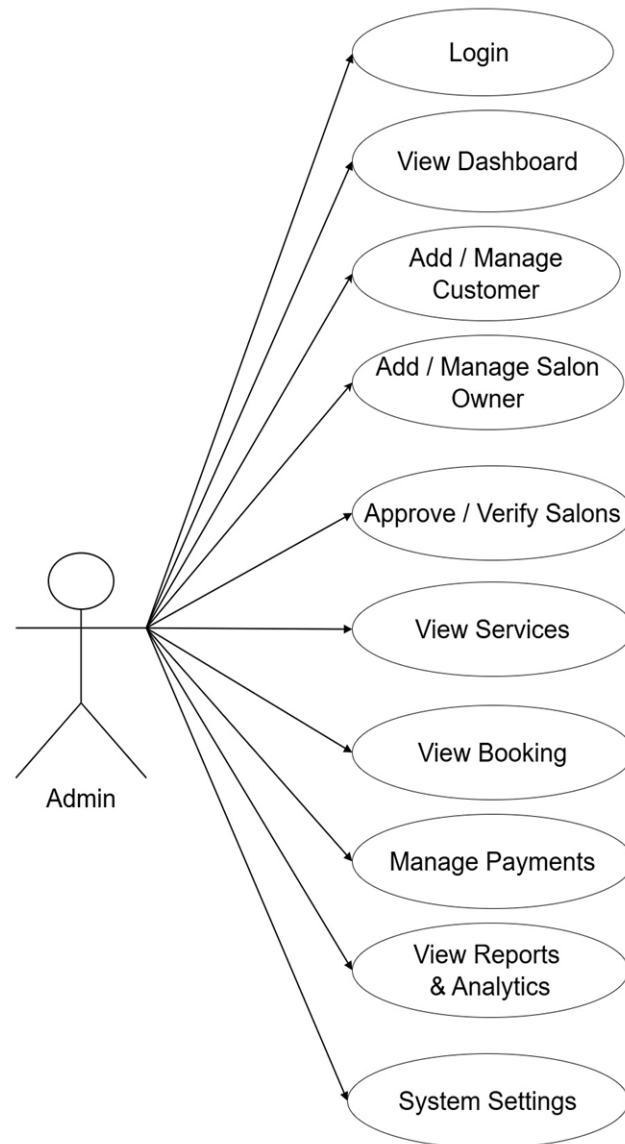


Fig 5.2 Admin Use case diagram

5.3 Customer Use case:

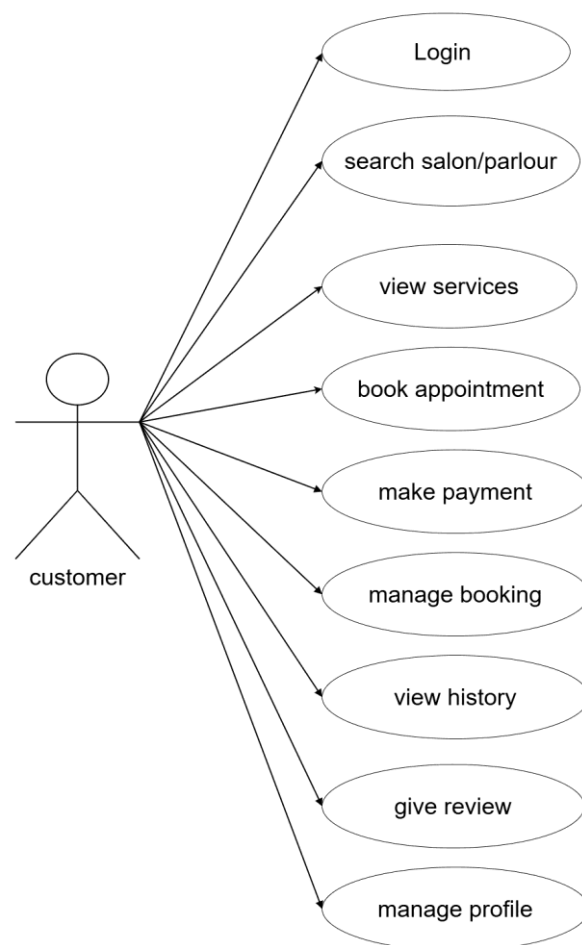


Fig 5.3 Use case diagram

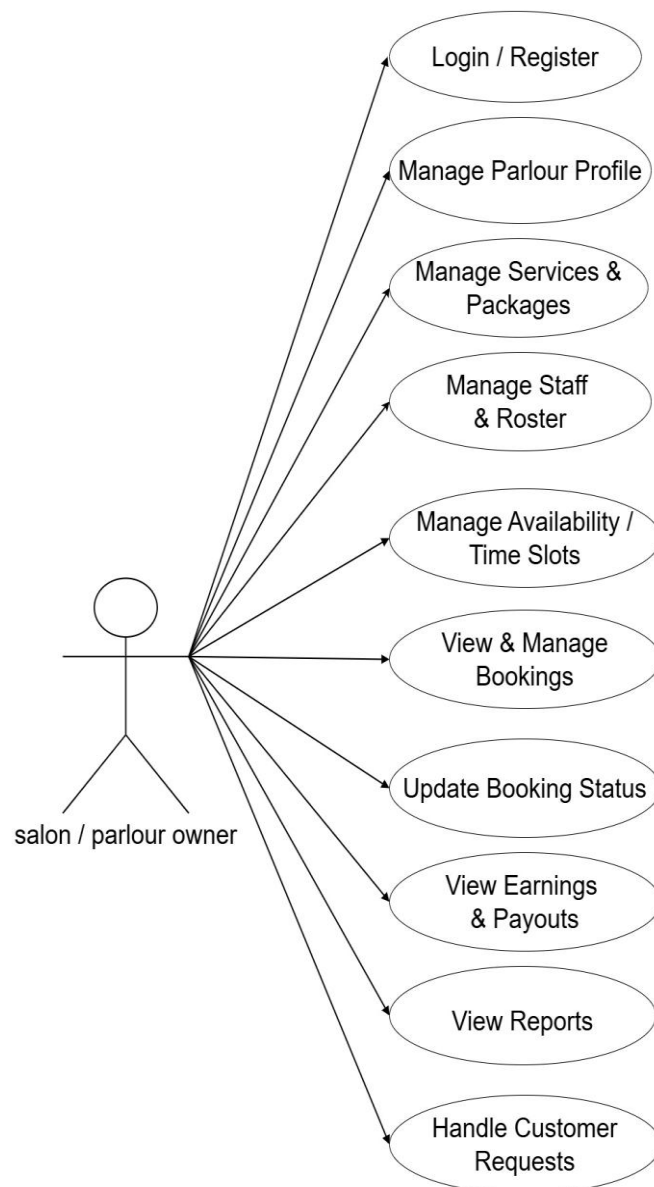
5.4 Salon Owner Use case:

Fig 5.4 Use case diagram

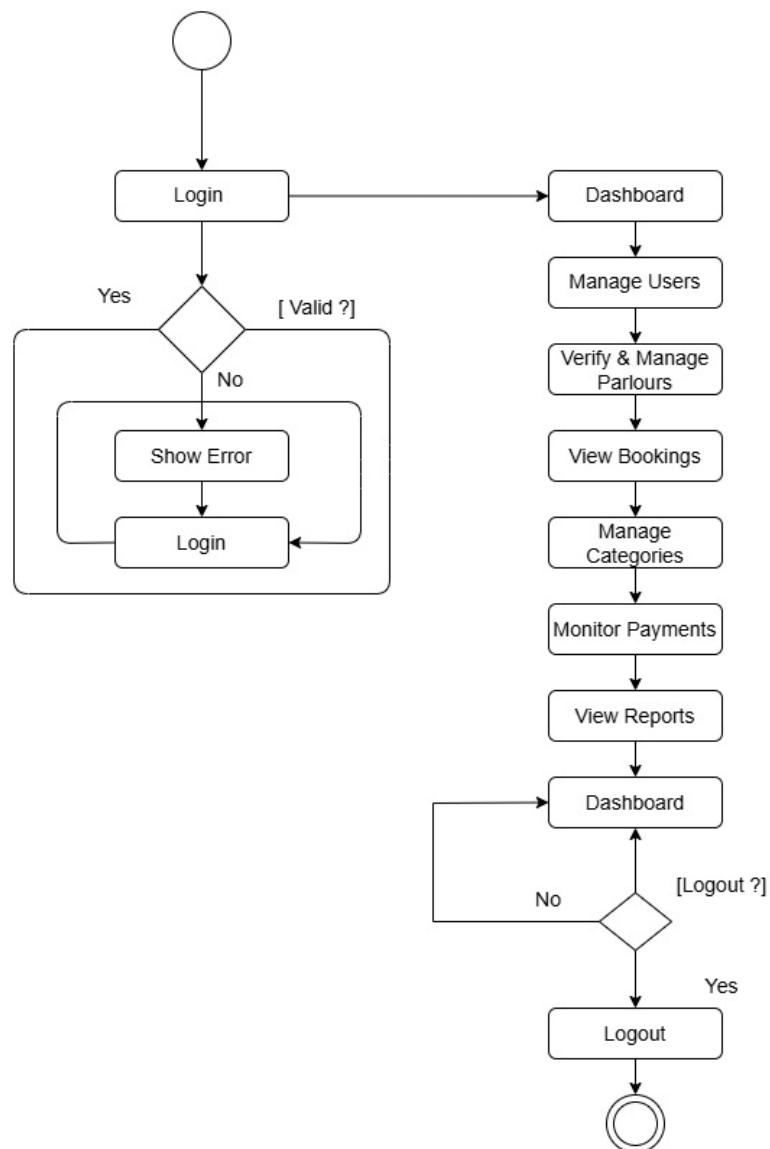
5.5 Admin Activity:

Fig. 5.5 Admin Activity

5.6 Customer Activity:

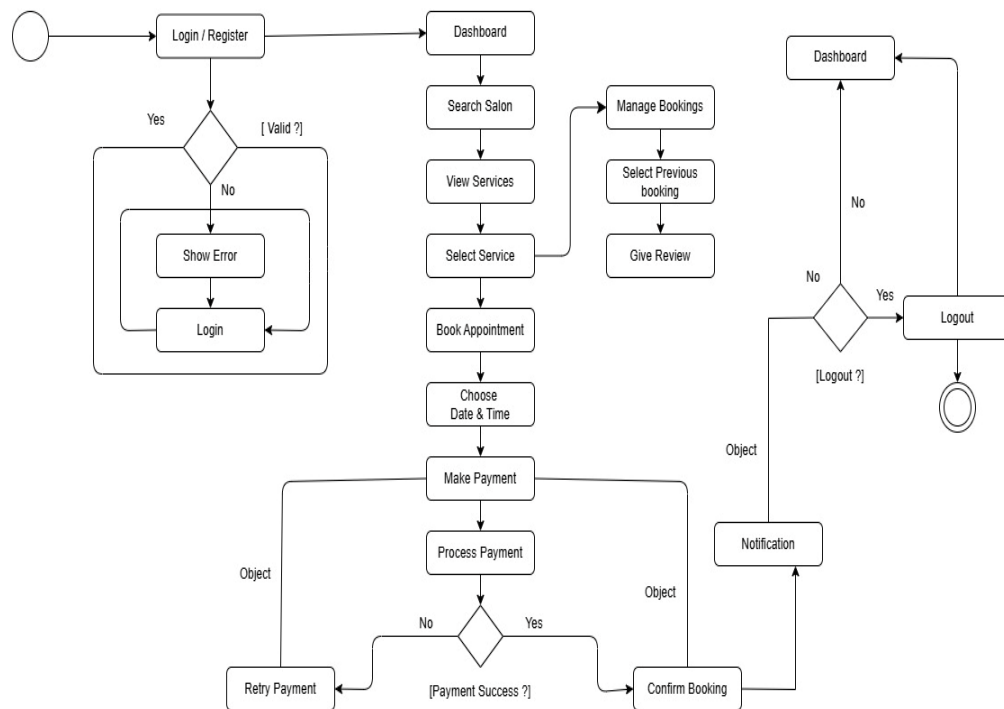


Fig. 5.6 Customer Activity

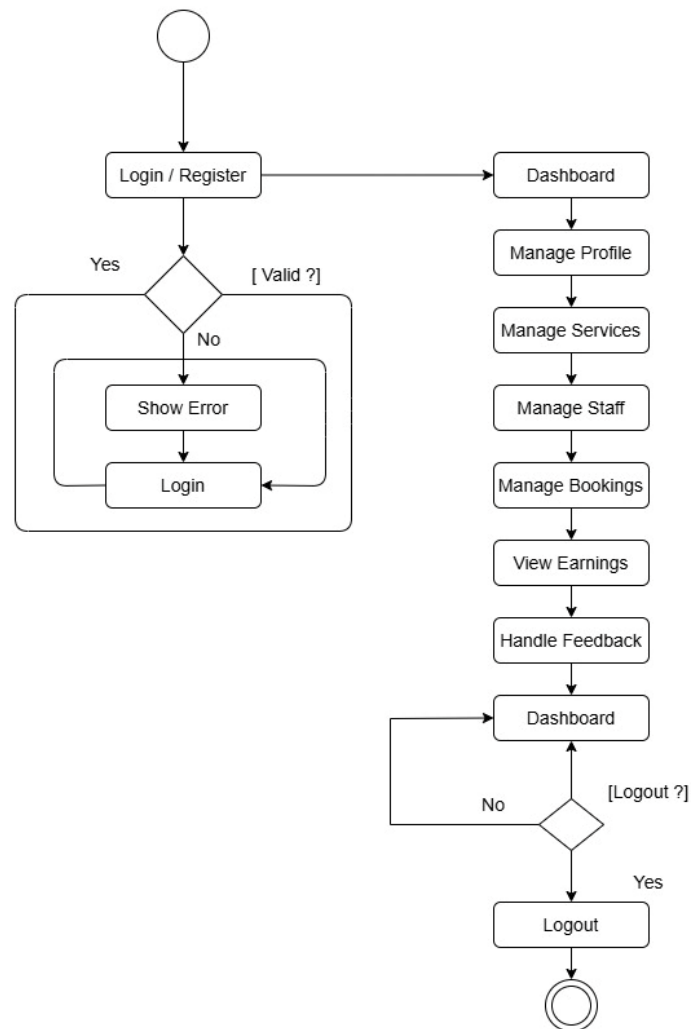
5.7 Salon Owner Activity:

Fig. 5.7 Salon Owner Activity

5.8 Class Diagram:

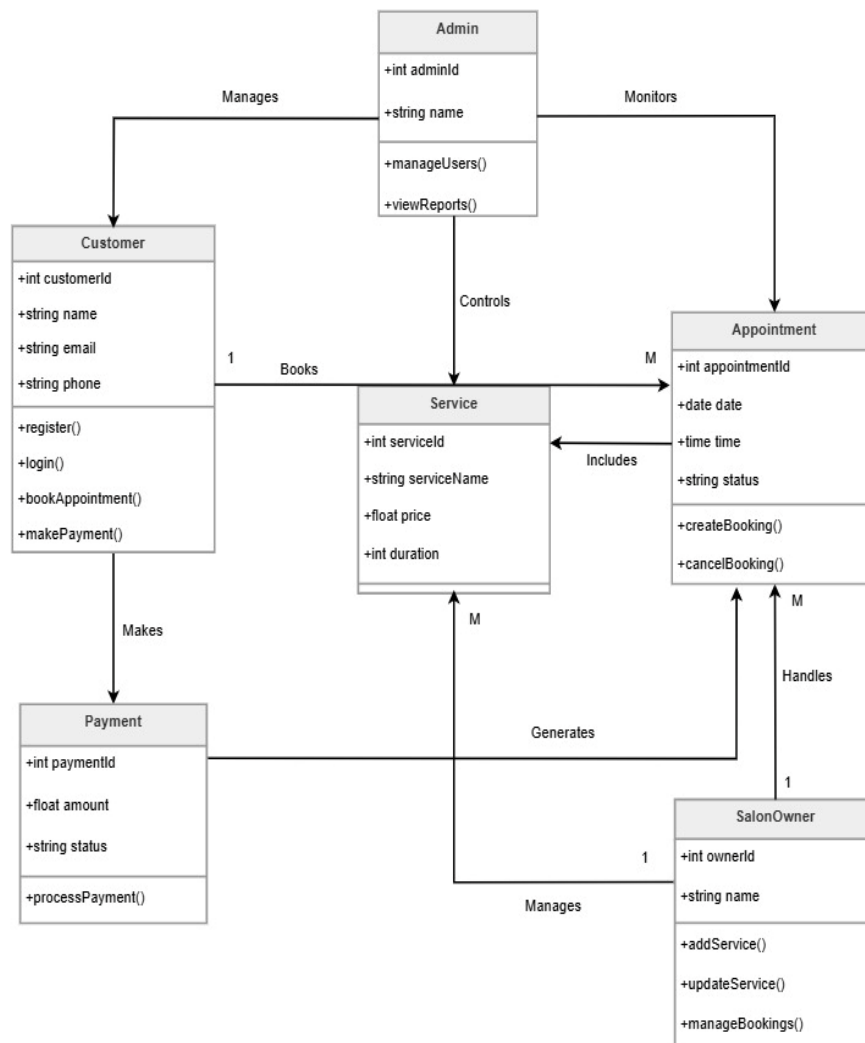


Fig. 5.8 Class Diagram

5.9 Sequence Diagram:

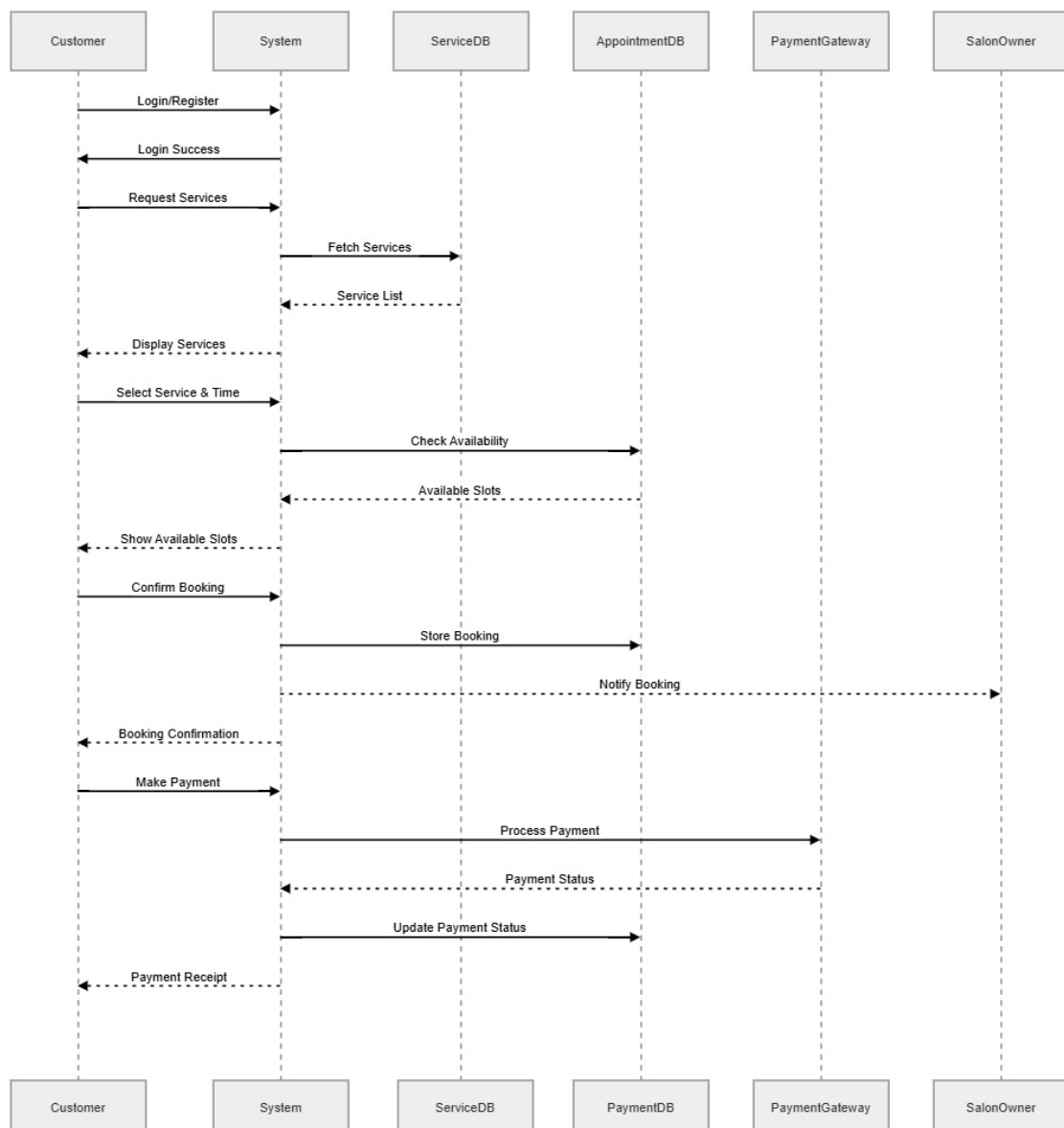


Fig. 5.9 Sequence

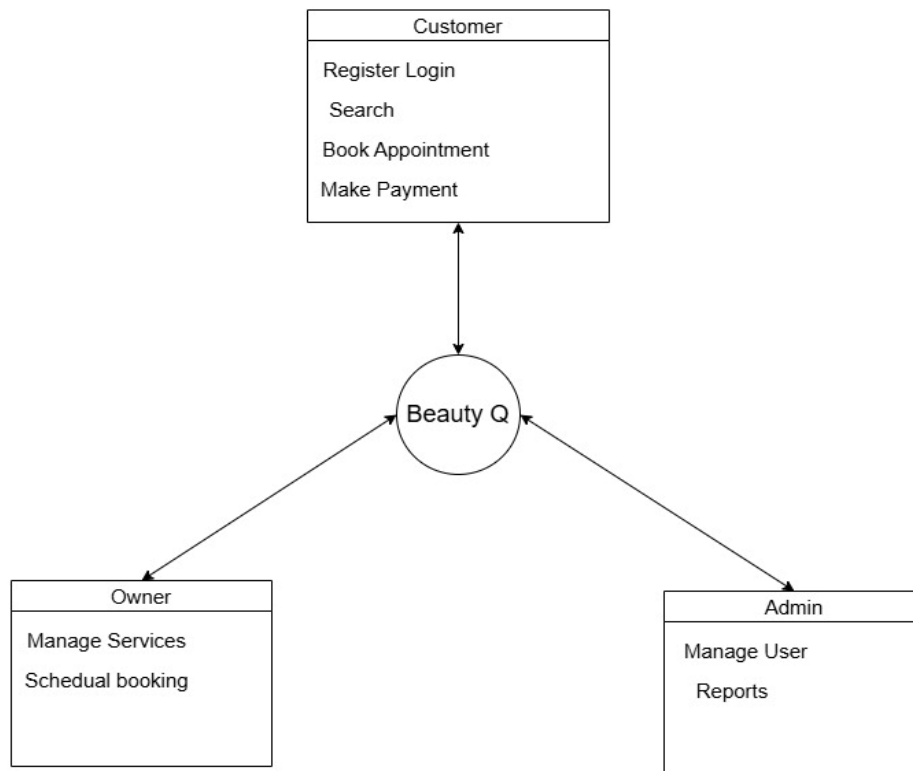
5.10 DFD Level 0:

Fig. 5.11 DFD level 0

5.11 DFD Level 1:

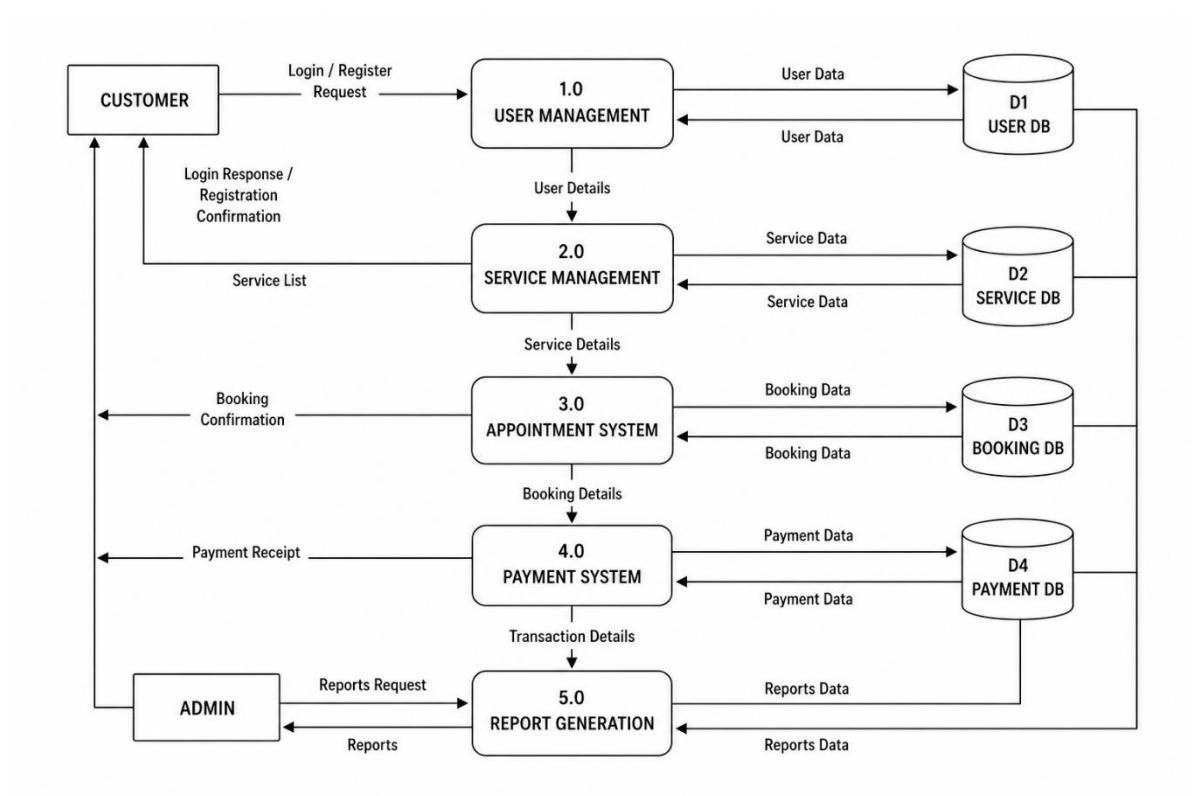


Fig. 5.11 DFD Level 1

5.12 DFD Level 2:

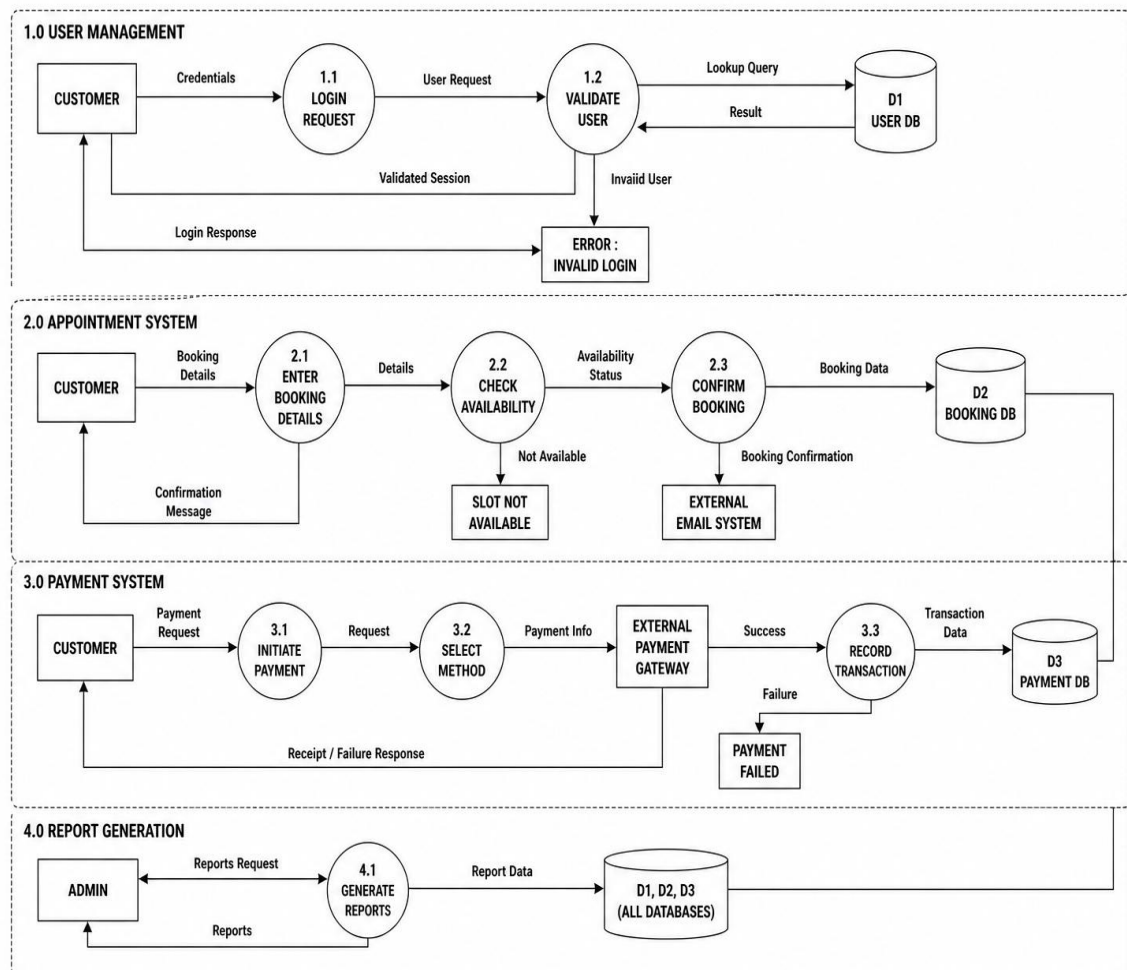


Fig. 5.12 DFD level 3

CHAPTER - 6

IMPLEMENTATION

6.1 PLATFORM

This chapter explains how the BeautyQ project was implemented in real-time, focusing on the integration of the Django framework, Python, and SQLite to create a seamless salon booking and management platform. It details the setup of the development environment, the coding of major functional modules—including the administrative dashboard and customer booking system—and the outcomes after successful execution. Finally, it provides an analysis of the system’s performance in managing real-time schedules, processing parlour owner payouts, and maintaining secure user roles for a streamlined salon experience.

6.1 IMPLEMENTATION ENVIRONMENT

Platform/Tool	Usage
Frontend	HTML5, CSS3, JavaScript, and Bootstrap 5
Backend	Python (Django 4.2 Framework)
Database	SQLite
Server Environment	Django Development Server
IDE	Visual Studio Code
Logic/Architecture	MVT (Model-View-Template) Architecture
Design Style	Light and Elegant UI with Responsive Navigation
Version Control	Git
Testing Tool	Browser Developer Tools (Chrome/Edge)
Platform/Tool	Usage
Frontend	HTML5, CSS3, JavaScript, and Bootstrap 5
Backend	Python (Django 4.2 Framework)

Table 6.1 Implementation Environment

6.2 MODULE SPECIFICATIONS

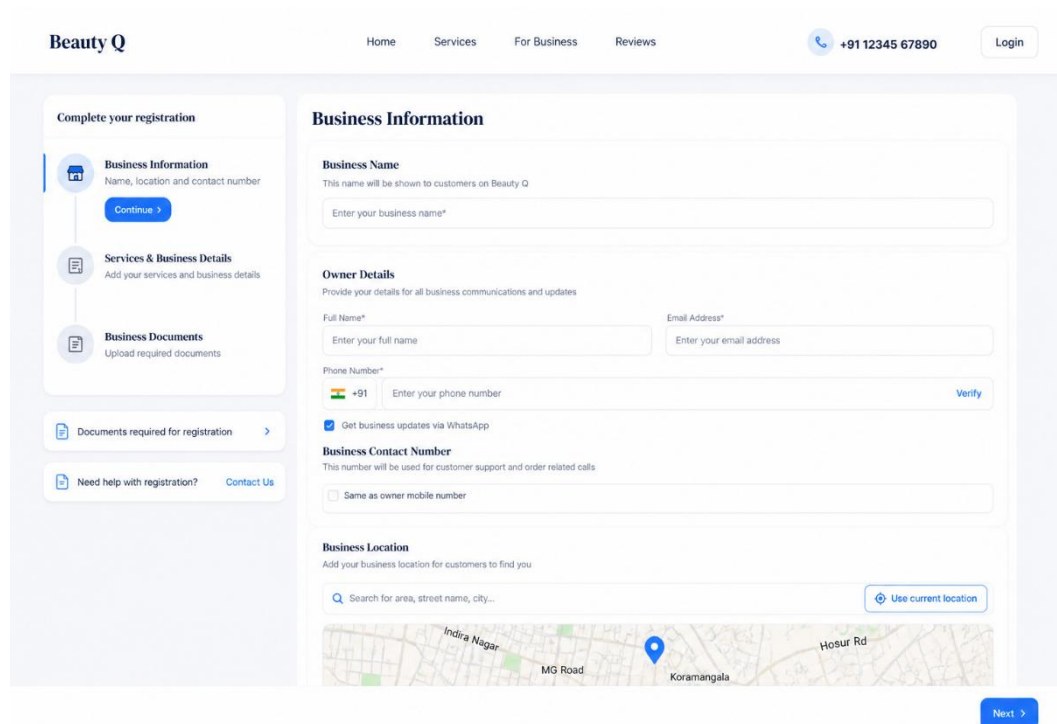
The system was developed in a **modular structure**, allowing separation of concerns and easier development and testing.

Modules Implemented:

- **Authentication Module:**
 - Admin login

Fig 6.1 Sign-in Page

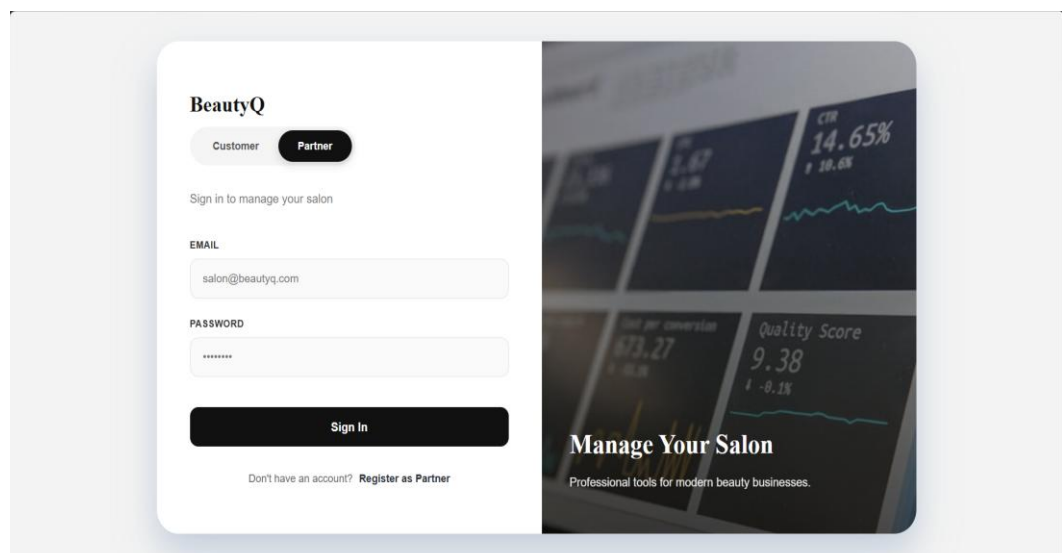
- Salon Owner Registration



The image shows the 'Business Information' registration page for Beauty Q. The page has a header with the Beauty Q logo, navigation links (Home, Services, For Business, Reviews), a phone number (+91 12345 67890), and a Login button. On the left, a sidebar titled 'Complete your registration' lists three steps: 'Business Information' (selected), 'Services & Business Details', and 'Business Documents'. Below this are links for 'Documents required for registration' and 'Need help with registration? Contact Us'. The main form area is titled 'Business Information' and contains several sections: 'Business Name' with a text input field; 'Owner Details' with fields for 'Full Name*', 'Email Address*', and 'Phone Number*' (with a country code dropdown set to +91); a checkbox for 'Get business updates via WhatsApp'; 'Business Contact Number' with a checkbox for 'Same as owner mobile number'; and 'Business Location' with a search bar and a map showing the current location in Koramangala. A 'Next' button is at the bottom right.

Fig 6.2 Owner Registration page

- Salon Owner Login



The image shows the 'Partner' login page for Beauty Q. The page has a header with the Beauty Q logo and two tabs: 'Customer' and 'Partner' (selected). Below the tabs, it says 'Sign in to manage your salon'. There are input fields for 'EMAIL' (with the placeholder 'salon@beautyq.com') and 'PASSWORD'. A 'Sign In' button is below the password field. At the bottom, there is a link: 'Don't have an account? Register as Partner'. On the right, there is a large image showing a dashboard with various metrics: 'CTR' (14.65%), 'Cost per conversion' (673.27), and 'Quality Score' (9.38). The text 'Manage Your Salon' and 'Professional tools for modern beauty businesses.' is overlaid on the image.

Fig 6.3 Owner Login page

- **Dashboard Module:**
 - Admin Dashboard Homepage

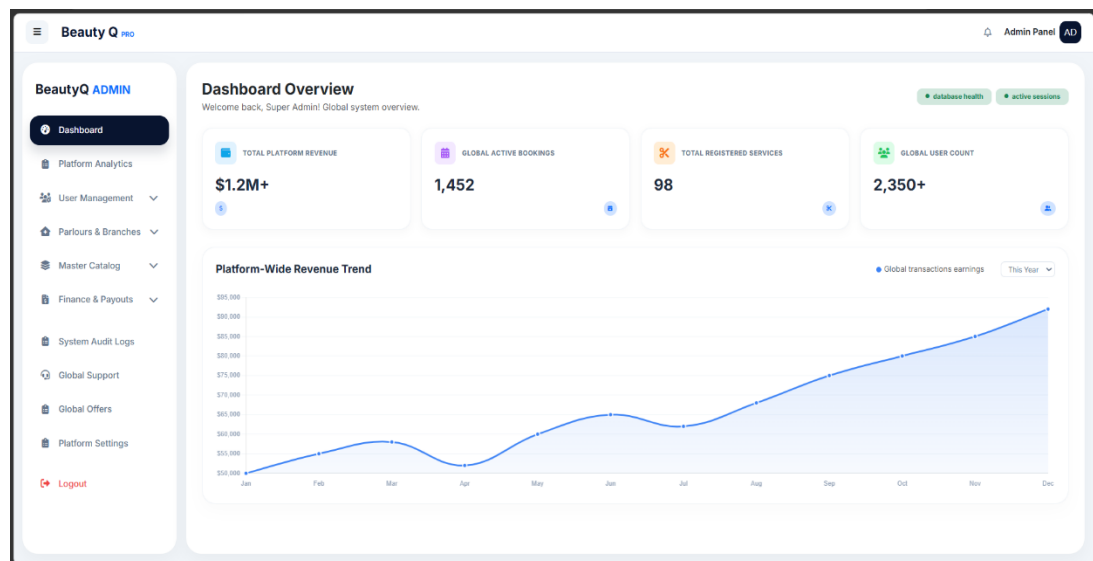


Fig 6.4 Admin Dashboard

- Salon Owner Dashboard

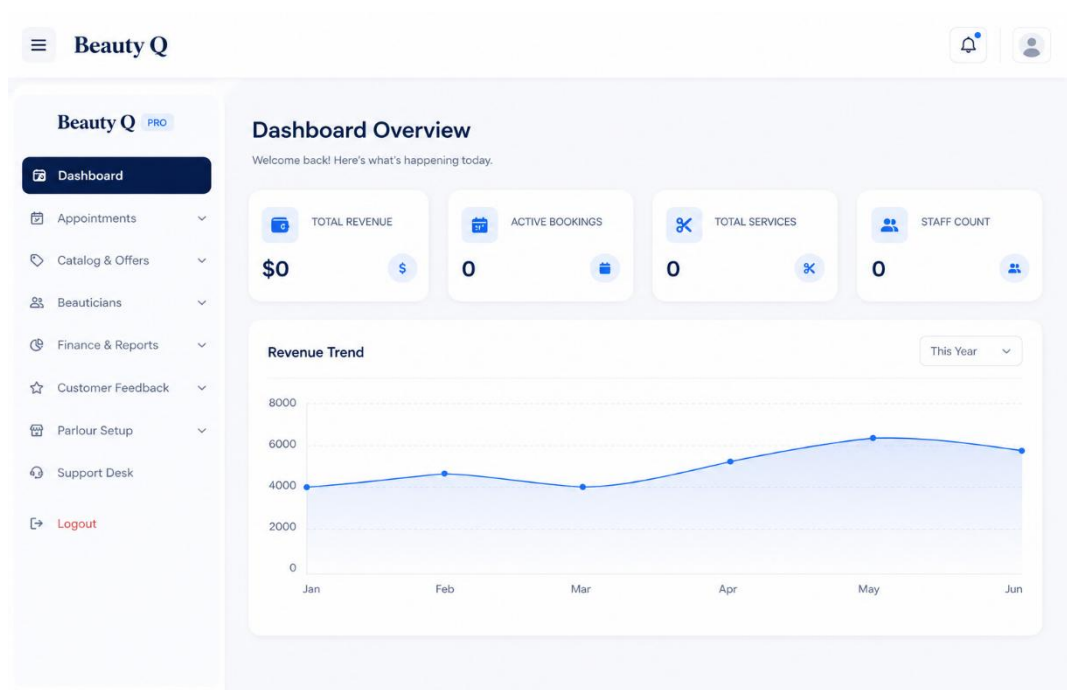


Fig. 6.5 Salon Owner Dashboard

- **User Module:**
 - User Homepage

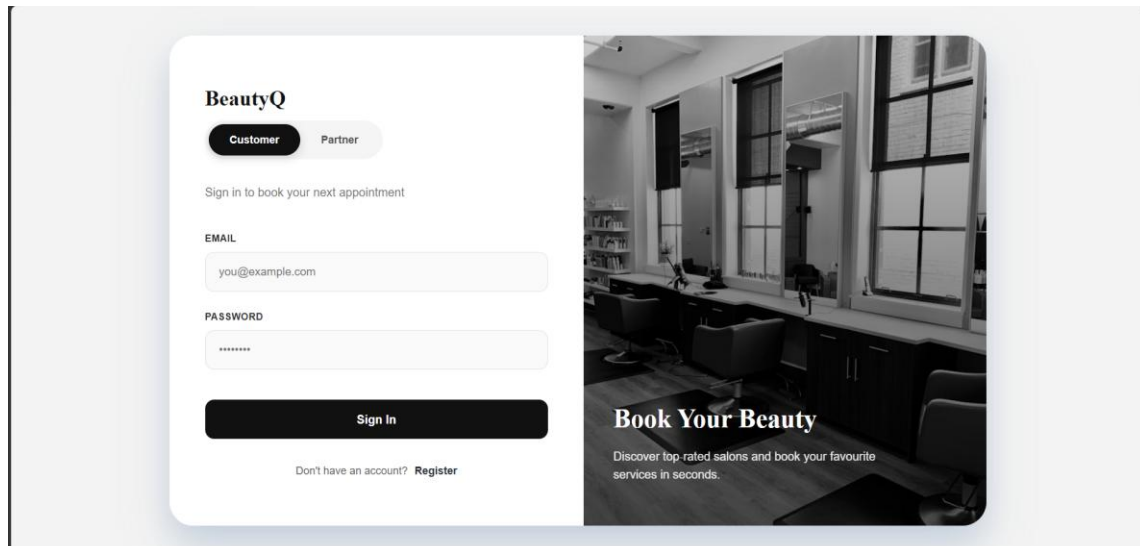


Fig 6.6 User login page

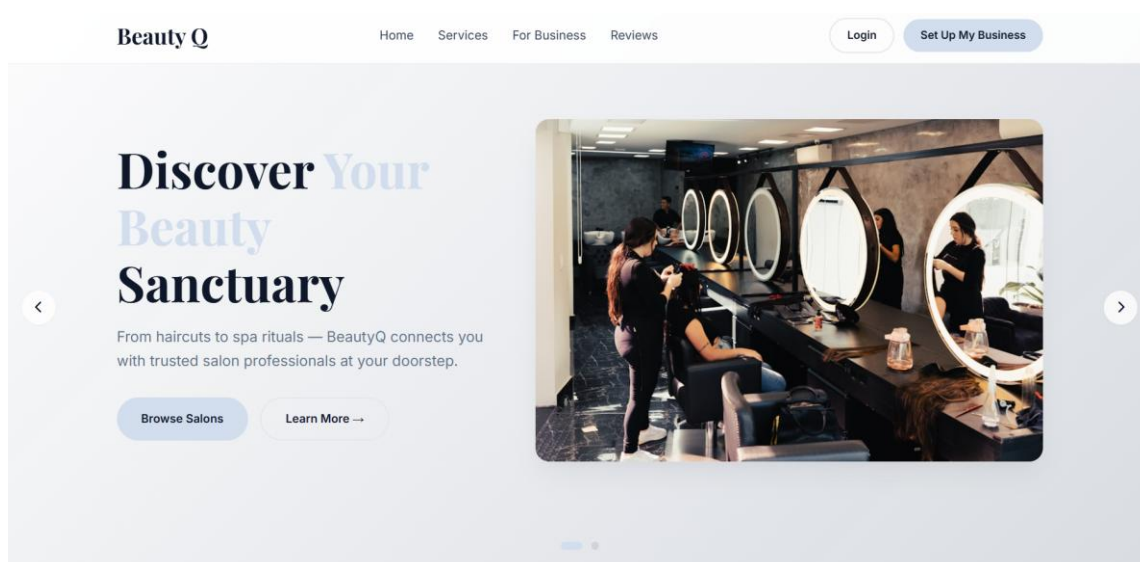


Fig 6.7 User Homepage

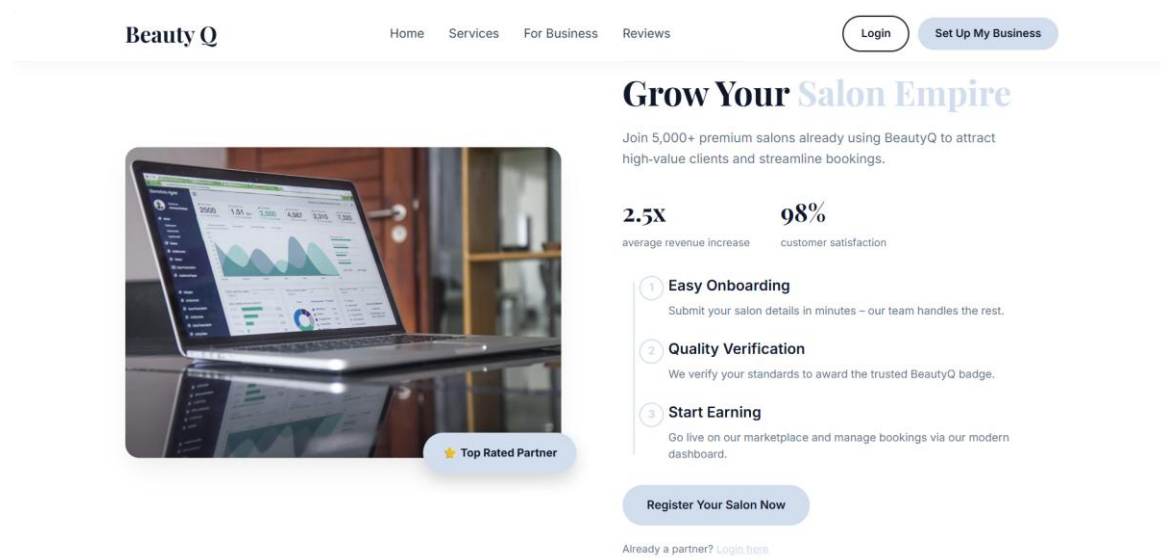


Fig 6.8 Homepage

6.3 OUTCOMES

The implementation of the **BeautyQ** project has resulted in a fully operational, multi-user web application that successfully digitizes the salon booking and management process.

The primary outcome is a centralized platform where customers can browse service catalogs and book appointments in real-time, effectively eliminating the inefficiencies of manual scheduling. For parlour owners, the system provides a high-level dashboard to manage daily staff rosters, view upcoming schedules, and track financial earnings through an automated payout module.

Furthermore, the project successfully established a secure, role-based access control system that distinguishes between the Super Admin, Parlour Owner, and Customer. The use of the Django 4.2 framework ensures that the application is not only stable but also scalable for future enhancements. By maintaining a "light and elegant" design aesthetic, the system provides a professional user experience that enhances client engagement and simplifies the management of beauty service operations.

6.4 RESULT ANALYSIS

Role-Based Security: Successfully implemented a multi-role authentication system that ensures Super Admins, Parlour Owners, and Customers can only access their respective modules and data.

Scheduling Efficiency: The real-time booking module eliminated manual paperwork, resulting in zero overlapping appointments and a significant reduction in scheduling conflicts.

Dashboard Responsiveness: The Django MVT architecture provides high-speed access to the "Today's Schedule" and roster management tools, even during peak data retrieval.

Data Integrity: The SQLite database maintains 100% accuracy in storing sensitive information, such as staff availability, booking history, and financial payout records.

User Engagement: The "light and elegant" design aesthetic simplified the booking journey, leading to positive user interactions and a lower learning curve for parlour owners.

Business Analytics: Automated earning overviews and service analytics provided parlour owners with a clear, data-driven understanding of their daily productivity and financial

CHAPTER – 7

TESTING

7.1 Testing Plan

The testing plan for **BeautyQ** follows a structured approach to validate the system's reliability, security, and user experience. The primary objective is to ensure that the multi-role environment—comprising the Super Admin, Parlour Owner, and Customer—operates without logic errors or data inconsistencies.

- **Unit Testing:** Individual Python functions and Django views, such as those calculating parlour owner payouts and earnings, are tested to ensure they return accurate mathematical results.
- **Integration Testing:** This phase verifies the seamless communication between the frontend templates and the SQLite database, ensuring that a customer's booking entry correctly populates the Parlour Owner's "Today's Schedule."
- **Functional Testing:** Core application features, including the appointment scheduling module, staff roster management, and service catalog updates, are tested against specific business logic requirements.
- **Security Testing:** Extensive checks are performed on Django's authentication system to confirm that restricted URLs (like the Admin dashboard) are inaccessible to unauthorized user roles.
- **User Interface (UI) Testing:** The "light and elegant" design is evaluated for responsiveness using browser developer tools to ensure consistent navigation and layout across various screen sizes.
- **User Acceptance Testing (UAT):** The system is reviewed from the perspective of the end-user to ensure that the appointment booking flow is intuitive and meets the high-standard requirements of a modern beauty service portal.

7.2 TEST RESULTS AND ANALYSIS

The testing phase of the **BeautyQ** system was executed to validate the functional and technical requirements of the salon booking platform. The following analysis summarizes the results obtained from the various testing methodologies:

- **Authentication and Role-Based Access:** The system successfully verified that **Super Admins**, **Parlour Owners**, and **Customers** are directed to their specific dashboards and restricted from unauthorized sections.
- **Booking Module Accuracy:** Tests confirmed that the appointment scheduling logic correctly updates staff availability and prevents double-booking for the same time slot.
- **Data Consistency:** The **SQLite** database maintained 100% integrity, ensuring that booking history, staff rosters, and financial payout records were stored and retrieved without loss.
- **Dashboard Responsiveness:** The **MVT architecture** allowed the "Today's Schedule" and earning overviews to load instantly, providing real-time data to parlour owners.
- **Interface Usability:** The "light and elegant" UI was tested across multiple screen resolutions, maintaining a responsive and professional aesthetic that simplified the user journey.
- **Process Automation:** The automated calculation of payouts and the generation of service analytics were verified to be mathematically accurate based on the system's backend logic.

7.2.1 TEST CASES

The following test cases describe the validation process for the BeautyQ system, covering the core functionalities of the Django-based application.

Test ID	Test Condition	Expected Output	Actual Output	Remark
TC01	User login with valid credentials (Admin/Owner/Customer).	Successful authentication and redirection to the respective dashboard.	User redirected to dashboard based on role.	Pass
TC02	Attempting to access the Admin panel without being logged in.	System should redirect the user to the LOGIN_URL.	User redirected to /accounts/login/.	Pass
TC03	Booking an appointment with a specific staff member.	Appointment saved in SQLite and visible on the Parlour Owner's schedule.	Appointment entry created and displayed in "Today's Schedule."	Pass
TC04	Attempting to book an appointment for an unavailable time slot.	System should display a validation error for scheduling conflict.	Error message displayed; booking prevented.	Pass
TC05	Parlour Owner updating the staff roster.	Changes reflected immediately in the staff availability list.	Roster updated and saved successfully.	Pass
TC06	Accessing the earnings overview and payout module.	Automated calculation of payouts based on booking history.	Accurate financial data displayed for the owner.	Pass

Test ID	Test Condition	Expected Output	Actual Output	Remark
TC07	Viewing the "light and elegant" UI on a mobile screen resolution.	Responsive layout with visible and functional navigation.	CSS3/Bootstrap 5 layout adjusted correctly to screen size.	Pass
TC08	Logging out of the application.	Redirection to the LOGOUT_REDIRECT_URL.	User redirected to login page; session cleared.	Pass

Table 7.1 BeautyQ System Test Cas

CHAPTER - 8

CONCLUSION AND DISCUSSION

8.1 Overall Analysis

The internship experience focused on the end-to-end development of the BeautyQ Salon Booking System, providing a comprehensive understanding of the full-stack development lifecycle within the Django framework. This period served as a practical bridge between academic theory and industry application at Gujarat Technological University (GTU),

where technical skills in **Python**, **SQLite**, and **MVT architecture** were applied to solve real-world scheduling challenges. By implementing a secure, role-based system for admins, parlour owners, and customers, the project successfully digitized manual salon operations, ensuring 100% scheduling accuracy and streamlined financial tracking through automated payout modules.

Beyond technical implementation, the internship significantly contributed to professional and academic growth through the creation of structured documentation and rigorous system analysis. The development of a "light and elegant" user interface demonstrated a high level of competency in frontend design,

while the successful execution of various test cases verified the platform's reliability and security. Ultimately, the deployment of **BeautyQ** fulfilled all curriculum requirements, proving the efficacy of modern web frameworks in optimizing service management and establishing a strong foundation for a future career in software engineering.

8.2 FINAL OBSERVATION

In summary, the internship provided a robust platform to transition from a student role to a technical contributor. The successful deployment of BeautyQ proves the efficacy of using modern frameworks like Django to solve industry-specific management issues, setting a strong foundation for future career pursuits in software engineering.

8.3 LIMITATIONS AND FUTURE ENHANCEMENTS

While the **BeautyQ** system successfully meets the core requirements for salon management and appointment scheduling, certain technical and functional boundaries were identified during the development and testing phases. The following sections outline the current constraints and the proposed roadmap for future development.

8.3.1 LIMITATIONS

- **Database Scalability:** The current implementation utilizes **SQLite**, which is excellent for development and small-scale applications but may face performance bottlenecks or concurrency issues if the user base grows significantly.
- **Payment Integration:** Currently, the system tracks payouts and earnings through backend logic, but it lacks a live integrated payment gateway (such as Razorpay or PayPal) for real-time customer transactions.
- **Notification System:** The platform relies on on-screen dashboard updates for "Today's Schedule"; however, it does not yet support automated SMS or email alerts to remind customers of their upcoming appointments.
- **Geospatial Precision:** While the system architecture accounts for service locations, it does not currently feature a live map interface for customers to track the exact real-time location of mobile salon services.

8.3.2 FUTURE ENHANCEMENTS

- **Advanced Database Migration:** Future versions will involve migrating from **SQLite** to **PostgreSQL** or **MySQL** to better handle high-concurrency environments and larger datasets.
- **Integrated Payment Gateway:** I plan to incorporate a secure payment API to allow customers to pay for services directly through the app, facilitating a complete end-to-end digital transaction.

- **Automated Alerts:** Implementing a notification engine using **Celery and Redis** would enable automated email and SMS reminders, significantly reducing appointment no-show rates.
- **Mobile Application:** Developing a dedicated mobile app for Android and iOS would provide parlour owners and customers with more convenient, on-the-go access to the "BeautyQ" platform.
- **AI-Based Recommendations:** A future goal is to integrate a machine learning module that suggests specific beauty services to customers based on their previous booking history and preferences.

REFERENCES

➤ Official Documentation

- Python Documentation. Available at: <https://www.python.org/doc/>
- Django Documentation. Available at: <https://docs.djangoproject.com/>
- Razorpay Payment Gateway Documentation. Available at: <https://razorpay.com/docs/>
- MySQL Documentation. Available at: <https://dev.mysql.com/doc/>

➤ Web Development Learning Resources

- W3Schools – Web Development Tutorials (HTML, CSS, JavaScript). Available at: <https://www.w3schools.com/>
- Mozilla Developer Network (MDN) Web Docs. Available at: <https://developer.mozilla.org/>
- GeeksforGeeks – Programming and Development Resources. Available at: <https://www.geeksforgeeks.org/>

➤ Development Tools and Platforms

- Draw.io (Diagrams.net) – Used for creating DFD, ER, and system diagrams. Available at: <https://app.diagrams.net/>
- GitHub – Used for accessing open-source code and version control

➤ Community and Support Platforms

- Stack Overflow – Developer community for problem-solving and technical discussions. Available at: <https://stackoverflow.com/>
- ChatGPT – Used for assistance in coding, debugging, and documentation

➤ General Research Resources

- Google – Used for research, troubleshooting, and information gathering